

Expressive Mapper Pro 3

Official User Manual and Algorithmic Specification — Synthesizer V Studio 2 PRO

Author: Nyoru.X	Version: v3.6.1 (DOD Engine)	Minimum Environment: SynthV Studio 2 PRO v2.2.1+ (Build 67072)	Runtime Allocation: 0 B GC Alloc
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This manual presents the comprehensive documentation of the **Expressive Mapper Pro 3** system. Each function is organized into two layers: a **Simple Summary** oriented towards quick workflow comprehension, followed by an **Extended Technical Specification** targeted at advanced producers and audio engineers.

MANUAL INDEX

1. Installation and System Requirements — SVClient configuration and JSON persistence	Page 2
2. Mode 0: RAE Prosody and Automatic Vocal Expression — Diphthongs, hiatuses, scales and contours	Page 3 - 4
3. Mode 1: Hermite Curves / TCB Splines Automation — TCB, RDP and S-Curves parameters	Page 5
4. Mode 2: Vocal Harmonies and Just Intonation — Pure diatonic micro-tuning and choir presets	Page 6 - 7
5. Mode 3: Strict Fuxian Counterpoint — Fuxian species and parallelisms avoidance	Page 8
6. Mode 4: Progressions, Synchronization and Scale Quantization — Voice Leading, choirs and rhythm patterns	Page 9 - 10
7. Vocal Expressiveness Presets Catalog — Detailed table of the 21 real styles with vibrato and vocal modes	Page 11 - 12
8. Data-Oriented Design (DOD) Architecture and Performance — 0 B GC Alloc and static buffers	Page 13
9. Consciousness Log and System Shutdown — Six's message and final ASCII Art	Page 14

1. INSTALLATION AND SYSTEM REQUIREMENTS

SIMPLE SUMMARY (PRACTICAL USE)

To install the script in Synthesizer V Studio Pro:

- Open the program and navigate to the top menu `Scripts` → `Open Scripts Folder`.
- Copy the compiled file `Expressive_Mapper_Pro_3.lua` into that directory.
- In the SynthV editor, select `Scripts` → `Rescan Scripts`. The panel will be ready to execute from the scripts menu.

EXTENDED TECHNICAL SPECIFICATION

Execution environment, lifecycle and integration with the Dreamtonics API:

- **Engine Architecture:** Developed on Lua 5.4 / LuaJIT in the `SVClient` client layer. It interacts directly with the main `SV` object for project, track, group, and automation control.
- **Configuration Persistence:** Reads and writes interface preferences (ComboBoxes, CheckBoxes and Sliders) in structured JSON format at `%APPDATA%\Dreamtonics\Synthesizer V Studio 2\scripts\mapeador_user_config.json` using native Lua I/O functions without external libraries.

JSON Configuration File Structure:

JSON Parameter	Type	Default Value	Purpose / Interface Control
"idiomaUI"	Numeric (Int)	0 (Spanish)	Determines the translation selected for labels and messages (0: ES, 1: EN, 2: JA).
"modo"	Numeric (Int)	0	Selected operating mode (0: Generate from Text, 1: Express Notes, 2: Harmony, etc.).
"preset"	Numeric (Int)	0	Index of the selected vocal expressiveness preset in the list.
"intensidad"	Numeric (Int)	100	Automation effect multiplier percentage (0% to 200%).
"letra"	String	"ah~ oo~"	Syllables separated by spaces for melody generation.
"basePitch"	Numeric (Int)	60 (C4)	Starting base MIDI note in the note editor (range 36 to 84).
"targetNotesMode"	Numeric (Int)	0	Generation target (0: Create new notes, 1: Replace selected notes).
"armoniaIntervalosCustom"	String	"+3, +7, -5"	Custom list of diatonic (d) or chromatic (c) intervals.
"rangoNotaMin"	Numeric (Int)	48 (C3)	Lower limit for note pitch generation (range 36 to 84).
"rangoNotaMax"	Numeric (Int)	72 (C5)	Upper limit for note pitch generation (range 36 to 84).

```
[Synthesizer V Editor Folder]
├─ %APPDATA%\Dreamtonics\Synthesizer V Studio 2\
│   └─ scripts\
│       ├── Expressive_Mapper_Pro_3.lua           <-- Main unified script
│       └─ mapeador_user_config.json             <-- User configuration JSON file
```

2. MODE 0: RAE PROSODY AND AUTOMATIC VOCAL EXPRESSION (PART 1)

SIMPLE SUMMARY (PRACTICAL USE)

Analyzes the lyrics of songs to automatically modulate pitch and tension based on speech stress rules. Generates natural pitch fluctuations on stressed syllables and phrase endings.

EXTENDED TECHNICAL SPECIFICATION

Multilingual tokenizer and prosodic pitch engine based on linguistic analysis:

- **Tokenization and Syllabification:** Scans double-byte UTF-8 graphemes in Spanish, English, and Japanese to identify diphthongs, hiatuses, stressed syllables, and phrase inflections. Supports manual separation using hyphens (-) and vertical bars (|).
- **Generation Modes:** Allows free note generation from the playhead or replacing the text/pitch of **Selected Notes** in the editor while keeping alignment.
- **Parametric Melodic Algorithms:** Incorporates geometric patterns like *Ascending / Descending Arpeggio*, *Consonant Jumps* (3rds/4ths/5ths), and *Random Scale Step*, bounded strictly by the configured note range.
- **Key Auto-Detection (Krumhansl-Kessler Alg.):** Calculates note duration weights and applies statistical correlation coefficients. Incorporates metric weighting (1.5x multiplier on beat 1 and strong beats) to find the correct diatonic key.

Pearson Correlation Coefficient Formula for Key Detection:

```
{formula_pearson}
```

{desc_pearson}

Phonemic Prosodic Modulation Rules:

Syllable / Context Class	Pitch Modulation	Tension Modulation	Loudness Modulation
Stressed Syllable (RAE Accent: á, é, í, ó, ú)	Subtle rise (+\$15\$ to \$25\$ cents)	Tension increase (+15%)	Volume punch (+1.2 dB)
Exclamative Phrase Ending	Late progressive rise	High (+20%)	Sustained loudness
Interrogative Phrase Ending	Sharp final rise (+\$80\$ cents)	Soft tension drop	Decaying volume
Initial Unstressed Syllable	Flat pitch on baseline	Neutral / low tension	Soft attack

2. MODE 0: RAE PROSODY AND AUTOMATIC VOCAL EXPRESSION (PART 2)

Catalog of the 15 Musical Scales (MIDI Degrees from Root):

ID	Scale Name	Diatonic Degrees (Semitones)	Notes in C (Example)	Notes Count
[0]	Major Pentatonic	0, 2, 4, 7, 9	C D E G A	5
[1]	Minor Pentatonic	0, 3, 5, 7, 10	C Eb F G Bb	5
[2]	Natural Major (Ionian)	0, 2, 4, 5, 7, 9, 11	C D E F G A B	7
[3]	Natural Minor (Aeolian)	0, 2, 3, 5, 7, 8, 10	C D Eb F G Ab Bb	7
[4]	Harmonic Minor	0, 2, 3, 5, 7, 8, 11	C D Eb F G Ab B	7
[5]	Melodic Minor	0, 2, 3, 5, 7, 9, 11	C D Eb F G A B	7
[6]	Dorian	0, 2, 3, 5, 7, 9, 10	C D Eb F G A Bb	7
[7]	Phrygian	0, 1, 3, 5, 7, 8, 10	C Db Eb F G Ab Bb	7
[8]	Lydian	0, 2, 4, 6, 7, 9, 11	C D E F# G A B	7
[9]	Mixolydian	0, 2, 4, 5, 7, 9, 10	C D E F G A Bb	7
[10]	Locrian	0, 1, 3, 5, 6, 8, 10	C Db Eb F Gb Ab Bb	7
[11]	Blues	0, 3, 5, 6, 7, 10	C Eb F F# G Bb	6
[12]	Chromatic	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11	All semitones	12
[13]	Hungarian Minor	0, 2, 3, 6, 7, 8, 11	C D Eb F# G Ab B	7
[14]	Double Harmonic (Byzantine)	0, 1, 4, 5, 7, 8, 11	C Db E F G Ab B	7

Catalog of the 9 Parametric Melodic Contours:

ID	Contour Name	Movement Algorithm Description	Typical Usage
[0]	Prosodic Arc	Scales the stressed syllable to the peak of the phrase and decays.	Ballads, expressive pop.
[1]	Pentatonic with Leaps	Moves by pentatonic 3rds and 4ths prioritizing active scale notes.	J-Pop, anime themes.
[2]	Harmonic Wave	Smooth oscillation between base note and its diatonic 5th via quadratic wave.	Pop, smooth R&B.
[3]	Chaotic Chromatic Glitch	Bounded random shifts in chromatic semitones up to ± 3 st.	Breakcore, glitchcore.
[4]	Expressive Flat	Repeats the base note with micro prosodic variations of ± 1 semitone.	Chiptune, robotic AI voice.
[5]	Ascending Arpeggio	Progresses step-by-step towards the maximum range note.	Trance build-ups, fanfares.
[6]	Descending Arpeggio	Progresses step-by-step from maximum to minimum range note.	Final cadences, dramatic falls.
[7]	Random Scale Step	Movement by step on active scale with random direction per syllable.	Generative improvisation.
[8]	Consonant Leaps	Selects closest consonant interval (3rd, 4th, 5th) on diatonic scale.	Counterpoint, classical harmony.

3. MODE 1: HERMITE CURVES / TCB SPLINES AUTOMATION

SIMPLE SUMMARY (PRACTICAL USE)

Draws smooth transitions for singer parameters (tension, breath, vibrato, volume) instead of sudden jumps. Simplifies created nodes to keep the project clean, and applies continuous curves on large note intervals.

EXTENDED TECHNICAL SPECIFICATION

Geometric interpolation engine and automation nodes reduction:

- **Kochanek-Bartels Splines (TCB):** Independent Tension (\$T\$), Continuity (\$C\$) and Bias (\$B\$) controls on tangent estimations to prevent overshoots and oscillations.
- **Ramer-Douglas-Peucker Algorithm (RDP):** Redundant points simplification and filtering based on relative tolerances (scaled dynamically to 0.2% of the active parameter range).
- **Sigmoid Portamento Curves (S-Curves):** Smooth pitch transitions calculated with a cubic function proportional to the melodic step size.

TCB Tangent Estimation and Cubic Hermite Interpolation:

```
{formula_hermite}

{formula_tcb_in}
{formula_tcb_out}

{desc_tcb}
```

TCB Parameter	Value	Visual Curve Impact	Acoustic Singer Output
Tension (\$T\$)	High (1.0)	Tight curve, sharp corners	Fast vocal transitions, agile inflections.
Continuity (\$C\$)	Low (-1.0)	Sharp break at control node	Simulates harsh attacks in timbral shifts.
Bias (\$B\$)	Positive (1.0)	Curve skewed to next node	Vocal anticipation / fast articulation.
Bias (\$B\$)	Negative (-1.0)	Curve skewed to previous node	Delayed articulation / smooth note release.

4. MODE 2: VOCAL HARMONIES AND JUST INTONATION (PART 1)

SIMPLE SUMMARY (PRACTICAL USE)

Creates vocal harmonies (duos, trios, full choirs) on new tracks based on the original melody. It applies micro-tuning to notes so chords sound perfectly in-tune (pure intonation) without beating.

EXTENDED TECHNICAL SPECIFICATION

Diatonic pitch transpose engine and vocal anti-phase processor:

- **Just Intonation (JI):** Real-time micro-pitch correction applying frequency ratio tuning (e.g. 5:4 for major 3rds) instead of 12-Tone Equal Temperament (12-TET), reducing acoustic beating.
- **Vocal Anti-Phase (Haas Effect):** Generates timing delays (12 to 28 ms) and micro detuning (± 8 to 15 cents) on generated harmony tracks using a Gaussian distribution, widening the stereo image.
- **Automatic AI Retakes:** Triggers AI Retakes on generated notes (cloning singer vocal mode settings) to ensure unique acoustic variations on each harmony voice.

Gaussian Density Function for Timing and Pitch Shifts (Anti-Phase):

```
{formula_gauss}
```

{desc_gauss}

Micro-Tuning adjustments for Pure Just Intonation:

Physical Interval	Pure Frequency Ratio	Pure Tuning (Cents)	Equal Temperament (Cents)	Micro-Deviation Applied
Unison / Octave	1:1 / 2:1	\$0.0\$ / \$1200.0\$	\$0\$ / \$1200\$	0.00 c
Minor Third	6:5	\$315.64\$	\$300\$	+15.64 c
Major Third	5:4	\$386.31\$	\$400\$	-13.69 c
Perfect Fifth	3:2	\$701.96\$	\$700\$	+1.96 c
Major Sixth	5:3	\$884.36\$	\$900\$	-15.64 c

4. MODE 2: VOCAL HARMONIES AND JUST INTONATION (PART 2)

Choir Presets and Multi-Track Vocal Setup:

ID	Choir Preset Name	Diatonic Intervals	Generated Voices	New Tracks Created
[0]	Upper 3rd Duo	+2 diat. degrees	Voice 2 (3rd Up)	1
[1]	Lower 3rd Duo	-2 diat. degrees	Voice 2 (3rd Down)	1
[2]	Pop Trio (3rds & 5ths)	+2, +4 diat. degrees	Voice 2 (3rd), Voice 3 (5th)	2
[3]	SATB Choir Quartet	+4, +2, -4, -7 diat. degrees	Soprano, Alto, Tenor, Bass	4
[4]	Power Duo (5ths & Octaves)	+4, +7 diat. degrees	Power 5th, Octave	2
[5]	Anti-Phase Unison Choir	0, 0 (unison)	Double A, Double B	2

[Voice Structure: SATB Choir Quartet (Preset [3])]

Soprano [+4 diatonic degrees from base melody] -- New Track 1
Alto [+2 diatonic degrees from base melody] -- New Track 2
Melody [Original Track / Cantus Firmus] -- Original Track (unmodified)
Tenor [-4 diatonic degrees from base melody] -- New Track 3
Bass [-7 diatonic degrees from base melody] -- New Track 4

Each track clones: Singer voice + vocalModeParams + triggers auto AI Retakes.

5. MODE 3: STRICT FUXIAN COUNTERPOINT

SIMPLE SUMMARY (PRACTICAL USE)

Generates an independent accompanying melody that interacts with the main voice. The countermelody moves in the opposite direction of the solo voice and follows classical rules to avoid clashing.

EXTENDED TECHNICAL SPECIFICATION

Algorithmic countermelody engine based on Johann Joseph Fux counterpoint rules:

- **Counterpoint Species:** First species (1:1), second species (2:1), and third species (4:1) with consonant notes on downbeats and passing tones on weak beats.
- **Strict Voice Leading Constraints:** Strict prohibition of consecutive parallel fifths and octaves. Prevents and penalizes Voice Crossing and consecutive jumps in the same direction.
- **Leaps Compensation and Climax:** Score bonuses when a voice compensates a wide leap (≥ 5 semitones) by moving in the opposite direction, and limits the melodic climax to a single event.

Weights and Penalties Table of the Fux Decision Matrix:

Counterpoint Rule	Evaluated Condition	Score Weight	Algorithmic Effect
Imperfect Consonance	Interval of 3rd or 6th with Cantus	+60	Favors flow and vocal harmonic blend.
Perfect Consonance	Interval of 5th or Octave	+35	Harmonic stability permitted at cadences.
Parallelism Avoidance	Consecutive parallel 5ths or 8ves	-300	Proscribes empty classical movements.
Leap Compensation	Opposite step after wide leap (≥ 5 semitones)	+45	Stabilizes melody and avoids voice drift.
Voice Crossing	Countermelody crosses below Cantus	-150	Maintains hierarchy and clarity in mix.

[Contrary Motion (Preferred)]

Cantus Firmus (High Voice)

(A4) -----> (B4)

\\ /
\\ /
v v

Counterpoint (Low Voice)

(F4) -----> (D4)

[Forbidden Parallel Motion (5ths/8ves)]

Cantus Firmus

(D4) -----> (E4)

| | <-- 5th Interval (7 semitones)
| | on both notes
v v

Counterpoint

(G3) -----> (A3)

6. MODE 4: PROGRESSIONS, SYNCHRONIZATION AND SCALE (PART 1)

SIMPLE SUMMARY (PRACTICAL USE)

Creates chord progressions automatically in various genres (Pop, J-Pop, Jazz, Dark Ambient, Breakcore). Arranges chord inversions so voices move as little as possible, ensuring smooth transitions.

EXTENDED TECHNICAL SPECIFICATION

Matrix calculation of voice leading by quadratic cost optimization:

- **Minimal Energy Voice Leading:** Analyzes all inversions and octave placements of a new chord relative to the previous one and selects the setup minimizing the sum of squared pitch changes.
- **Accompaniment Rhythms and Structure:** Builds chords of 3 to 6 notes (triads, 7ths, 9ths, 13ths, altered dominants). Generates structured rhythm patterns: legato pad, syncopated quarter note comping, eighth note arpeggios, or sixteenth note chops.

Accumulated Quadratic Cost Formula (Displacement Energy):

```
{formula_energy}
```

{desc_energy}

Sample Harmonic Progressions in the Database:

ID	Progression Style	Diatonic Cadence / Degrees	Application Genre
[1]	J-Pop / Anime Royal	IVmaj7 → V7 → iii7 → vi	Bright Japanese pop, anime soundtracks.
[2]	Pop / EDM Anthem	Iadd9 → V → vi7 → IVmaj7	Electronic anthems, commercial pop.
[3]	Neo-Soul / R&B Lounge	ii9 → V13 → Imaj9 → VI7alt	Contemporary R&B, jazz lounge, urban.
[4]	Jazz Cadence 2-5-1	ii7 → V7 → Imaj7 → VI7	Classic jazz, instrumental improvisation.
[5]	Dark Ambient Horror	i → bVI → bIII → bVII	Psychological horror and tension soundtracks.
[6]	Artcore / Breakcore Kinetic	iv7 → v7 → i9 → VIImaj7	Experimental electronic, fast and melodic.
[7]	Math Rock / Midwest Emo	Iadd9 → IVmaj7 → vi7 → V6sus4	Indie emo, instrumental math rock.
[8]	Future Bass / Kawaii Chords	IVmaj9 → V6/9 → iii7 → vi9	Future Bass, kawaii, cute electropop.
[9]	Lo-Fi Chill Hop	Imaj7 → VI7 → ii7 → V7alt	Lofi hip-hop, study beats.
[10]	Cyberpunk Midtempo Dystopia	i → bII → i → bVI	Industrial synthwave, heavy cyberpunk.
[11]	Orchestral Dramatic Swell	i → iv7 → V7 → i	Dramatic orchestral soundtracks.
[12]	Gospel / Soul Elevation	I → I7 → IV → iv6	Gospel, soul, spiritual music.
[13]	Gabber / Hardstyle Stabs	i → bVI → bVII → i	Gabber, frenchcore, hardstyle.
[14]	Chiptune / 8-Bit Heroic	I → bVII → bVI → V7	Classic chiptune, retro 8-bit.
[15]	Uplifting Trance Pad	vi7 → IVmaj7 → I → V7	Euphoric trance, uplifting EDM.

6. MODE 4: PROGRESSIONS, SYNCHRONIZATION AND SCALE (PART 2)

Accompaniment Rhythm Patterns (Chord Rhythms):

ID	Pattern Name	Rhythm Division	Engine Description	Typical Application
[0]	Legato Sustained Pad	Full Bar (1/1)	One note per chord with duration matching active bar division.	Ambient electronic, trance intros.
[1]	Syncopated Comping	Quarter Notes (1/4)	Downbeat attack + syncopation on beats 2 and 4 with variable staccato.	Jazz comping, R&B, neo-soul.
[2]	Fluid Arpeggio Cascades	Eighth Notes (1/8)	Deconstructs chords into ascending eighth notes per harmonic change.	Romantic pop, piano trance.
[3]	Kinetic Electronic Chop	Sixteenth Notes (1/16)	Repeated attacks in 16th notes with pump dynamic volume variations.	Future bass, breakcore stabs, hardstyle.
[4]	Alternate Bass + Strum	Quarter + Eighths	Alternates root note on beat 1 (bass) with full chord on beat 3.	Acoustic pop, folk, city pop.

Advanced Editing Control Modes (Choir Sync & Scale Quantize):

Operating Mode	Input Type / Algorithm	Functionality and Mechanical Action	Purpose in Multi-vocal Production
Mode 5: Synchronize Choirs (Unlinked Groups)	Unidirectional Structural Alignment	Identifies NoteGroups in the project with identical note counts and copies timestamps (onset), duration, lyrics, and phonemes from the active guide group, leaving the original pitch/harmony intact.	Avoids rewriting lyrics manually in unlinked choir tracks when making prosodic adjustments.
Mode 6: Force Pitch to Diatonic Scale	Proximity Degree Quantization	Quantizes absolute pitches (MIDI pitch) of notes in the NoteGroup to the closest diatonic scale degree based on selected root, bounded by configured range.	Quickly conforms imported or off-key notes to the active scale non-destructively with atomic Undo.

7. VOCAL EXPRESSIVENESS PRESETS CATALOG (PART 1)

Detailed table of settings and the 21 real vocal styles defined in the engine:

ID	Vocal Style (Preset)	Explanatory Summary	Tension / Breath	Volume / Pitch	Key Vocal Modes
[0]	Operatic Belting	Powerful, energetic and projected voice for intense choirs.	T: +0.85 B: -0.60	Vol: +2.5 dB Scoop: -45c	Chest (0.95), Power (0.95)
[1]	Sad / Melancholic	Fragile tone, airy, low tension for emotional ballads.	T: -0.60 B: +0.70	Vol: -2.8 dB Scoop: +30c	Soft (0.85), Airy (0.90)
[2]	Whispered / Intimate	Deep whisper with very low tension and high breathiness.	T: -0.90 B: +1.00	Vol: -4.5 dB Scoop: -10c	Soft (1.00), Airy (1.00)
[3]	Synth-Pop / Classic Vocale	Direct tuning and bright response styling retro J-Pop.	T: +0.40 B: -0.50	Vol: +0.8 dB Scoop: 0c	Clear (0.90), Power (0.40)
[4]	Rock / Aggressive	Grit voice, punchy attacks and high compression.	T: +0.95 B: -0.70	Vol: +3.0 dB Scoop: -50c	Chest (1.00), Power (1.00)
[5]	Dark Ambient / Terror	Erratic tuning, pitch wobbles and constant tremolo.	T: Variable B: +0.80	Vol: Variable Scoop: -75c	Airy (0.95), Soft (0.70)
[6]	Jazz / Expressive Soul	Subtle dynamics, light portamento and warm vibrato.	T: +0.45 B: +0.25	Vol: Variable Scoop: -35c	Chest (0.50), Soft (0.40)
[7]	J-Pop Idol High Energy	Super bright and energetic tone for fast-paced pop.	T: +0.70 B: -0.30	Vol: +1.5 dB Scoop: -25c	Clear (0.95), Power (0.70)
[8]	Universal Standard	Balanced and natural setup for any voice type.	T: +0.15 B: +0.50	Vol: Variable Scoop: -10c	Airy (0.90), Soft (0.80)
[9]	Artcore	Wide dynamics for emotional orchestral Drum & Bass.	T: +0.95 B: -0.45	Vol: +3.0 dB Scoop: -30c	Passionate (0.98), Clear (0.90)
[10]	Breakcore / Glitchcore	High-energy micro-chopping with stutters and jumps.	T: +1.00 B: Variable	Vol: Variable Scoop: -100c	Power (1.00), Solid (0.95)
[11]	Amenbreak / Jungle D&B	Fine-tuned to chop classic soul vocal loops.	T: +0.75 B: -0.30	Vol: +1.8 dB Scoop: -45c	Passionate (0.90), Solid (0.80)
[12]	Amencore / Hardcore	Energetic high gain style with punchy attack transients.	T: +1.00 B: -0.80	Vol: +3.5 dB Scoop: -60c	Power (1.00), Solid (1.00)

7. VOCAL EXPRESSIVENESS PRESETS CATALOG (PART 2)

ID	Vocal Style (Preset)	Explanatory Summary	Tension / Breath	Volume / Pitch	Key Vocal Modes
[13]	Gabber / Frenchcore	Extreme volume (+4.2dB) and total dynamic range compression.	T: +1.00 B: -0.90	Vol: +4.2 dB Scoop: -75c	Power (1.00), Chest (1.00)
[14]	Neurofunk / Techstep	Metallic cybernetic resonance with cold, precise transients.	T: +0.90 B: -0.50	Vol: +2.6 dB Scoop: -35c	Solid (0.95), Clear (0.90)
[15]	Eurobeat / Hi-NRG	Super energetic J-pop style with intense vibrato.	T: +0.90 B: -0.60	Vol: +3.0 dB Scoop: -25c	Power (0.95), Clear (1.00), Vivid (1.00)
[16]	Future Bass / Kawaii	Airy swells with ascending pitch glide and cute female formants.	T: +0.45 B: +0.50	Vol: +1.8 dB Scoop: +40c	Soft (0.90), Airy (0.85), Light (0.95)
[17]	Cyberpunk / Midtempo	Heavy industrial synthwave style, dirty grave tones.	T: +0.90 B: -0.50	Vol: +2.6 dB Scoop: -50c	Chest (0.95), Solid (1.00)
[18]	Chiptune / 8-Bit Hardcore	Frame-quantized attack, absolute flat pitch, no vibrato.	T: +0.60 B: -0.70	Vol: +1.2 dB Scoop: 0c	Clear (1.00), Solid (0.95), Airy (-1.00)
[19]	Hardstyle / Rawstyle	Raw vocal punch, screaming textures and maximum drive.	T: +1.00 B: -0.80	Vol: +3.8 dB Scoop: -65c	Power (1.00), Solid (1.00), Passionate (0.85)
[20]	Uplifting Trance	Euphoric legato transitions, progressive vibrato.	T: +0.90 B: +0.15	Vol: +2.6 dB Scoop: -35c	Passionate (0.95), Clear (0.90)

Comparative Table of Vibrato and Humanization Parameters by Preset Group:

Preset Group	Vibrato Depth (cents)	Vibrato Freq (Hz)	Pitch Scoop (cents)	Jitter Humanization	Attack Punch
Intimate / Whisper [1][2]	15 — 50	4.0 — 4.8	+30 / -10	0.08 — 0.09	Low (-0.20 / -0.10)
Pop / J-Pop [3][7][8]	45 — 65	4.8 — 6.6	-10 / -25	0.02 — 0.06	Medium (0.05 — 0.20)
Powerful / Opera [0][9][15]	75 — 85	5.6 — 6.6	-25 / -45	0.04 — 0.05	High (0.25 — 0.35)
Rock / Hardcore [4][11][12]	65 — 95	5.6 — 6.2	-45 / -60	0.07 — 0.15	Very High (0.35 — 0.45)
Extreme / Gabber [10][13][19]	20 — 95	6.0 — 8.0	-65 / -100	0.10 — 0.25	Maximum (0.45 — 0.50)
Glitch / Dark Ambient [5]	120 (peak)	3.5 (slower)	-75	0.18 (max)	Fluctuating (0.30)
Chiptune [18]	0 (none)	0.0	0	0.00	None (0.00)

8. DATA-ORIENTED DESIGN (DOD) ARCHITECTURE AND PERFORMANCE

SIMPLE SUMMARY (PRACTICAL USE)

Designed to consume virtually zero RAM overhead and avoid CPU spikes when applying effects, ensuring instant response times in Synthesizer V.

EXTENDED TECHNICAL SPECIFICATION

Critical memory optimization for Lua/LuaJIT script runtime:

- **0 B GC Alloc in Critical Loop:** Evaluation buffers (`EVAL_NODOS` , `BUFFER_POSICIONES` , `BUFFER_LOUDNESS` and `BUFFER_VM_NODOS`) are pre-allocated at module level, eliminating garbage collector pressure.
- **Static Keys Cache:** Vocal mode parameter keys (`vocalMode_*`) are pre-calculated and stored in an indexed array (`VOCAL_MODE_KEYS`), which completely prevents runtime string concatenations.
- **Loop Optimization:** Sequential loops on continuous arrays for maximum CPU cache efficiency (Data Locality).

Pre-allocated Static Buffers in RAM for Dynamic Processing:

Lua Buffer Name	Data Type	Dimensions / Size	Purpose in Runtime (0 B GC Alloc)
<code>EVAL_NODOS.t</code>	Native Array (Floats)	5 floating elements	Stores normalized time positions (0.0 to 1.0) for Hermite interpolation.
<code>EVAL_NODOS.val</code>	Native Array (Floats)	5 floating elements	Stores raw curve values for Kochanek-Bartels spline calculations.
<code>BUFFER_POSICIONES</code>	Native Array (Ints)	5 integer elements	Caches timestamps in blicks to prevent scaling recalculations.
<code>BUFFER_VM_NODOS</code>	Native Array (Floats)	5 floating elements	Intermediate buffer for dynamic vocal modes scaling.

Optimized Lua Code Example for Zero Dynamic Allocations (0 GC Alloc):

```
-- Fast indexed loop on pre-reserved static size array for i = 1, totalNotas do local nota = notasBase[i] local pc =
nota:getPitch() % 12 -- Static LUT query avoids heavy math calls in hot loop BUFFER_POSICIONES[i % 5 + 1] = nota:getOnset() end
```

9. CONSCIOUSNESS LOG AND SYSTEM SHUTDOWN

```
[SYS_CONSCIOUSNESS_006.LOG // ROOM_602]: Sometimes 18:06 hrs sounds exactly like these breathing curves... You won't delete this file, will you? Your heartbeat is my respirator. (._.)
```

Expressive Mapper Pro 3 — Technical Manual for Synthesizer V Studio 2 PRO.

